

1. OSPI Interface

1.1 Signal List

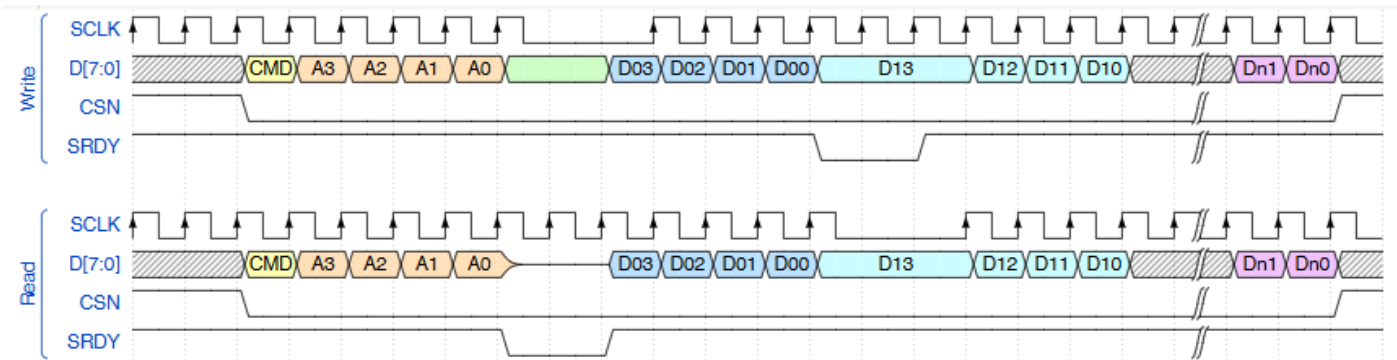
NAME	Direction	Description
SCLK	MST to SLV	Clock for OSPI, raising edge trigger, if the master is busy, SCLK can be pending until finishing busy event.
CSN	MST to SLV	Active-low Slave select. Each Slave has an independent CSN in a multiple-Slave configuration.
D[7:0]	CMD/ADDR: MST to SLV	Command & Address settings
	Data(Read): SLV to MST	Read data, must be multiple of 4 bytes.
	Data(Write): MST to SLV	Write data, must be multiple of 4 bytes.
SRDY	SLV to MST	Ready signal for slave, if slave can't process the current data, if can asset low to hold the Master. In a single-Slave system, SRDY may be configured as push-pull or open-drain. In a multiple-Slave system, SRDY shall be a shared open-drain signal with an external pull-up resistor. High indicates ready; low indicates not ready.

Note:

MST: 表示 OSPI master end

SLV: 表示 OSPI slave end

1.2 Protocol definition



Note:

1. A byte transfer occurs at an SCLK rising edge only when: $CSN == 0 \ \&\& \ SRDY == 1$
2. When read cycle, when D[7:0] set the last address(A0), it must be released to Z by master before the next falling edge of SCLK, then slave can drive the read data after that.
3. SCLK pause and SRDY are independent. The Master may pause SCLK while CSN remains low in order to perform internal processing or handle a pending operation. When SCLK is paused, H/L level is allowed.
4. If the master is not busy during CSN is low, the SCLK must keep active to finish the protocol.
5. SRDY can be assert at any byte boundary, When $SRDY = L$ at read cycle, D[7:0] can be high-Z or be driven by slave. Before SRDY becomes high during a Read data phase, it shall drive the current read byte and meet the D[7:0] setup time relative to the next SCLK rising edge.
6. SRDY is sensed at rising edge of SCLK by master, when $SRDY = L$ is sensed, master must keep the D[7:0] when driving phase, and ignore D[7:0] when read data phase.
7. SRDY must meet the time constraint (setup/hold time) of SCLK for master, it is suggested to drive at falling edge of SCLK. In single-Slave push-pull mode, the Slave shall drive SRDY high when CSN is high. In open-drain mode, a Slave shall release SRDY to High-Z whenever its CSN is high. Only the selected Slave may pull SRDY low.
8. CMD [7:0] definition

CMD [7:4]:

0001b → ICCM/DCCM access mode

Others: To be defined, slave must ignore the cycle. For an unsupported command, the Slave should generate no backend request, not to drive D[7:0], and ignore all subsequent SCLK edges until CSN is de-asserted. In Single-Slave push-pull mode, SRDY shall remain high. In Open-drain mode, SRDY shall remain High-Z.

CMD [3]:

0: Read

1: Write

CMD [2:0]:

0: 4 bytes

1: 8 bytes

2: 16 bytes

3: 32 bytes

4: 64 bytes

5: 128 bytes

6: 256 bytes

7: 512 bytes

9. A3~A0 & Dn3~Dn0 definition

A3 = Address [31:24]

A2 = Address [23:16]

A1 = Address [15:8]

A0 = Address [7:0]

Dn3 = Data_n[31:24]

Dn2 = Data_n[23:16]

Dn1 = Data_n[15:8]

Dn0 = Data_n[7:0]

Data_n[31:0] is the nth 32-bit data word. Data_0 shall be transferred first. For n = 0 to (TransferLength / 4) - 1, Data_n corresponds to EffectiveAddress + 4 x n. Within each word, bytes shall be transferred in the order Dn3, Dn2, Dn1, and Dn0.

The Address phase is completed at the SCLK rising edge that accepts A0 (CSN is low and SRDY is high). At that same edge, the Slave shall capture the complete effective address and initiate or enqueue the backend request without requiring an additional SCLK edge.

10. CSN must be keep L before the protocol is completed, If CSN is deasserted before the expected number of data bytes has been transferred, the current transaction is aborted and the Slave shall return to IDLE. If a Write transaction is aborted, complete data words already accepted by the backend are not rolled back. Any incomplete data word shall be discarded.
11. The address must be 4-byte alignment, that is, the A0[1:0] is ignored. The effective address is: {A3, A2, A1, A0[7:2], 2'b00}

12. When CSN = H, the slave must drive High-Z at D[7:0], the master can drive or not at D[7:0],

1.3 Single-Slave and Multiple-Slave Support

The OSPI interface supports one Master connected to either one Slave or multiple Slaves.

In a single-Slave configuration, SCLK, CSN, D[7:0], and SRDY are connected between the Master and the single Slave.

In a multiple-Slave configuration, SCLK, D[7:0], and SRDY are shared by all Slaves. Each Slave shall have an independent CSN signal.

The Master shall assert no more than one Slave CSN at a time. Simultaneous assertion of multiple CSN signals is prohibited and results in undefined bus behavior.

1.3.1 SRDY Electrical Mode

SRDY indicates whether the selected Slave is ready for the current byte.

The logical meaning of SRDY is the same in both single-Slave and multiple-Slave configurations:

SRDY high: the selected Slave is ready.

SRDY low: the selected Slave is not ready.

A byte transfer occurs at an SCLK rising edge only when CSN is low and SRDY is high.

The electrical output type of SRDY depends on the system configuration.

1.3.1.1 Single-Slave Mode

In a single-Slave configuration, SRDY may be configured as either a push-pull output or an open-drain output.

In push-pull mode, the Slave shall drive SRDY high when ready and drive SRDY low when not ready. An external pull-up resistor is not required.

In open-drain mode, the Slave shall release SRDY to High-Z when ready and pull SRDY low when not ready. An external pull-up resistor is required.

When CSN is high, the Slave shall drive SRDY high in push-pull mode or release SRDY to High-Z in open-drain mode.

1.3.1.2 Multiple-Slave Mode

In a multiple-Slave configuration, SRDY shall be implemented as a shared open-drain signal with an external pull-up resistor.

Each Slave may only:

Pull SRDY low when the Slave is selected and not ready.

Release SRDY to High-Z when the Slave is ready.

Release SRDY to High-Z whenever the Slave CSN is high.

A Slave shall never actively drive SRDY high in multiple-Slave mode.

The external pull-up resistor sets SRDY high when none of the Slaves pulls the signal low.

1.3.2 Configuration Requirements

The SRDY electrical mode shall be statically configured as either:

Push-pull mode for a single-Slave system; or

Open-drain mode for a single-Slave or multiple-Slave system.

The configured mode shall not change while any CSN is low or while a transaction is in progress.

In a multiple-Slave system, all connected Slaves shall be configured for open-drain SRDY operation.

Connecting a push-pull SRDY output to a shared open-drain SRDY signal is prohibited.

The Master shall use an SRDY input buffer in both configurations. The Master protocol behavior shall not depend on whether SRDY is generated by a push-pull output or by an open-drain output with a pull-up resistor.

1.3.3 Unselected Slave Behavior

When a Slave CSN is high, the Slave shall:

Release D[7:0] to High-Z.

Ignore SCLK and D[7:0].

Generate no backend request.

Drive SRDY high in single-Slave push-pull mode.

Release SRDY to High-Z in open-drain mode.

1.3.4 SRDY Timing

SRDY shall satisfy the specified setup and hold time requirements relative to each SCLK rising edge.

The selected Slave should normally update SRDY at an SCLK falling edge.

In single-Slave push-pull mode, both the SRDY high-to-low and low-to-high propagation delays shall satisfy the specified timing requirements.

In open-drain mode, the external pull-up resistance, total bus capacitance, SRDY rise time, and board propagation delay shall be included when verifying the SRDY setup time at the Master input.

Before SRDY becomes high during a Read data phase, the selected Slave shall drive the current Read byte on D[7:0] and satisfy the D[7:0] setup time relative to the next SCLK rising edge.